SASWATA MAITY

saswata.work.data@gmail.com
saswata-maity.github.io/research_portfolio +91-6291865918
GitHub: github.com/Saswata-Maity LinkedIn: linkedin.com/in/saswata-maity

Computer Vision & Deep Learning researcher with hands-on experience building and benchmarking architectures for medical image classification and segmentation. Three manuscripts in submission; open-source projects on Hugging Face and GitHub.

Work Experience

- **Specialist** — PwC, India
  Computer Vision & Intelligent Document Processing — *Jul 2026 – Present*
  - Developing an end-to-end OCR and document-recognition pipeline for invoices and business documents using PaddleOCR for text detection, TrOCR for recognition, and LayoutLMv3 for layout-aware key-information extraction.
  - Added confidence-based validation and human loop review, reducing manual verification effort by 40 %.
  - Built reproducible PyTorch training and evaluation workflows and field-level F1.

Experience

- **Research & Development Intern** — IIT Ropar, India
  Supervisor: Dr. Sukrit Gupta — Homi Bhabha Cancer Hospital & Research Centre, Punjab — *Jan 2026 – June 2026*
  - **Cervical Cancer Early Detection** | *PyTorch, Flask, FocalNet, Swin Transformer*
    - Curated a 700+ patient-case dataset from HBCH; addressed class imbalance by combining external datasets with a Focal-Tversky loss and auxiliary supervision.
    - Benchmarked CoAtNet, Swin, ResNet, and ConvNeXt for 4-class image classification; FocalNet achieved **89% accuracy, 94.92% AUC, ∼4% FPR**.
  - **Hospital-Based Cancer Registry Automation** | *RAG, Ollama, QLoRA, FastAPI, Pydantic, NLP*
    - Built a RAG pipeline with a QLoRA fine-tuned Ollama model for medical document understanding, served via FastAPI with Pydantic-validated structured outputs; automated retrieval, cleaning, and registry-format structuring of patient reports with human-in-the-loop validation.
    - Projected **∼60% improvement in operational efficiency** upon full deployment.
- **Analyst** — AIIMS Mohali, India
  Department of Academic Medicine — *Jan 2026 – June 2026*
  - Analyzed patient data for individuals with substance use disorders and performed geographic proximity analysis to identify optimal locations for new care centres spaced 5 km apart, improving access to timely care.
  - Performed data analysis and scripting on patient datasets and delivered interactive dashboards to support decision-making, directly contributing to the rollout of the new centres.

Research Works

- **Lightweight Plant Disease Classification**: Designed a **0.63M-parameter** CNN for mobile deployment using TensorFlow, Keras, and OpenCV, achieving **99.03% accuracy**; with Dr. Ram Sarkar, Jadavpur University (manuscript submitted).
- **Self-Supervised Medical Image Segmentation**: Developed a self-supervised segmentation approach for H&E-stained histopathology images; with Dr. Ram Sarkar, Jadavpur University (manuscript submitted).
- **Decomposition-Based Medical Image Segmentation**: Developed a decomposition-based segmentation approach for H&E-stained histopathology images; with Dr. Ram Sarkar, Jadavpur University.
- **Explainable AI for Histopathology Image Segmentation**: Developed a novel algorithm for explainable AI in histopathology images, efficiently training and extracting learned embeddings to provide clearer explanations of cancer cell segmentation; with Dr. Ram Sarkar, Jadavpur University (manuscript submitted).

Projects

- **CRA U-Net – Nuclei Segmentation (Hugging Face Space)**: Built a Channel Routed Attention U-Net (**7.5M params**) for nuclei segmentation using TensorFlow and Keras; achieved **85.23 Dice / 74.91 IoU** on MoNuSeg and **80.69 Dice / 67.77 IoU** on CryoNuSeg.
- **Additional Projects**: Rest of the projects available on GitHub: github.com/Saswata-Maity.

Education

- **Ramkrishna Mission College, Narendrapur (University of Calcutta)** — Narendrapur, India
  M.Sc. Data Science and Artificial Intelligence; CGPA: 8.49/10 — *Expected Jun 2026*
- **University of Calcutta** — Kolkata, India
  B.Sc. Computer Science; CGPA: 7.68/10 — *Aug 2024*

Skills

- **Languages**: Python
- **ML/DL Frameworks**: PyTorch, TensorFlow, Keras, Scikit-learn
- **Computer Vision**: OpenCV, image classification, semantic segmentation, transformer-based vision backbones (Swin, FocalNet, ConvNeXt, CoAtNet), U-Net architectures
- **Libraries**: NumPy, Pandas, Matplotlib, NLTK, Scrapy
- **Web & Tools**: Flask, FastAPI, SQL, Git
- **Domains**: Computer Vision, Medical Imaging, NLP, Generative AI, Bioinformatics

Certifications

- Jadavpur University – Data Science and AI (Offline); Grade: A — Apr 2025
- TensorFlow: Advanced Techniques Specialization – DeepLearning.AI (Coursera) — Mar 2025
- TensorFlow Developer Certificate – DeepLearning.AI (Coursera) — Mar 2025
- Deep Learning Specialization – DeepLearning.AI (Coursera) — Feb 2025
- Machine Learning Specialization – DeepLearning.AI (Coursera) — Apr 2024